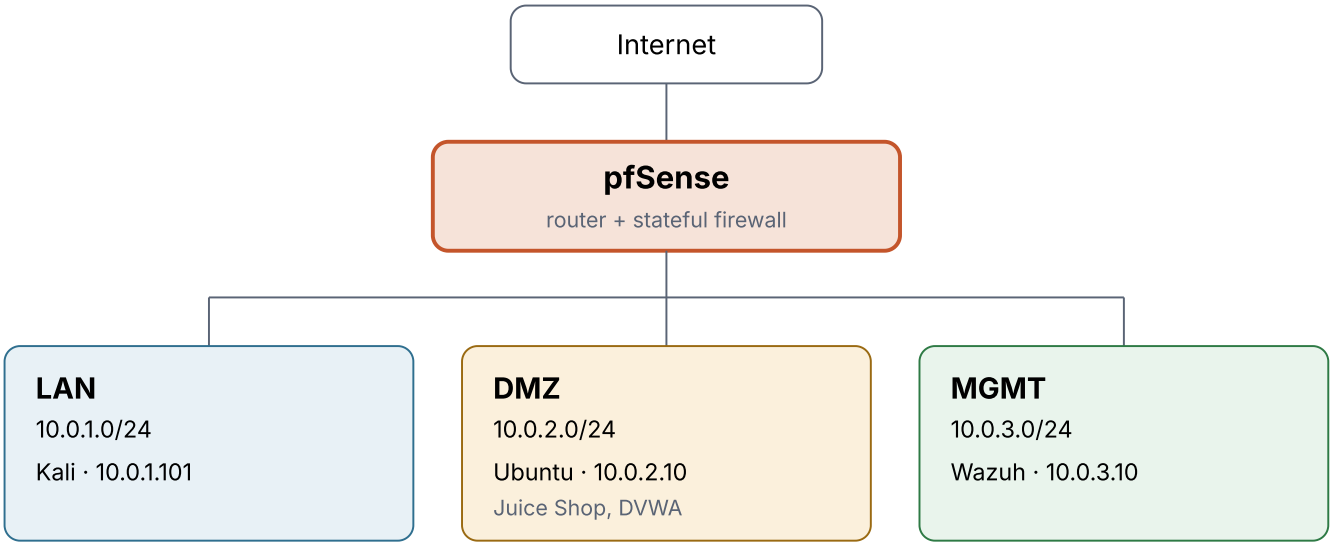
SOC HOME LAB · STUDY GUIDE

Networking Fundamentals you've already used

Every networking idea behind your pfSense, Kali, Ubuntu and Wazuh lab, explained simply with examples from your own network. Read it before the next lab session. Each chapter ends with a quick check.



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1 Networks, LANs and zones
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1 Networks, LANs and zones

> **A network is a group of devices that can send data to each other. Splitting one big network into smaller zones lets you control who talks to whom.**

LAN: Local Area Network

A **LAN** is a network in one place: a home, an office or a building. Devices on the same LAN can reach each other directly through a **switch**.

LAN segments in VMware

In VMware, a **LAN Segment** is a private virtual switch. Only the VMs you plug into it can talk on it. It has no link to your Windows laptop and no DHCP from VMware. Your lab has three: `soc-lab`, `soc-dmz` and `soc-mgmt`. They behave like three separate physical switches with their own cables.

The three zones

Zone	Full name	What lives there	Trust
LAN	Local Area Network	Users and analysts (Kali)	Medium
DMZ	Demilitarised Zone	Services exposed to attackers (web apps)	Low
MGMT	Management	Tools that watch and control everything (SIEM)	Highest

DMZ comes from the military: a strip of land between two hostile countries where neither side may keep troops. In networking, it's a **buffer** between the dangerous internet and your trusted network. If a DMZ server is hacked, the attacker is still stuck in the buffer.

MGMT is the most protected zone. Whoever controls it can see and change everything, including deleting the logs that would reveal them.

THINK OF IT LIKE THIS

Your lab is a building. The **LAN** is the open-plan office where staff work. The **DMZ** is the reception area where visitors are allowed in. **MGMT** is the locked security control room with the CCTV screens. **pfSense** is the guard at the doors between them, checking every person against a list.

IN YOUR LAB

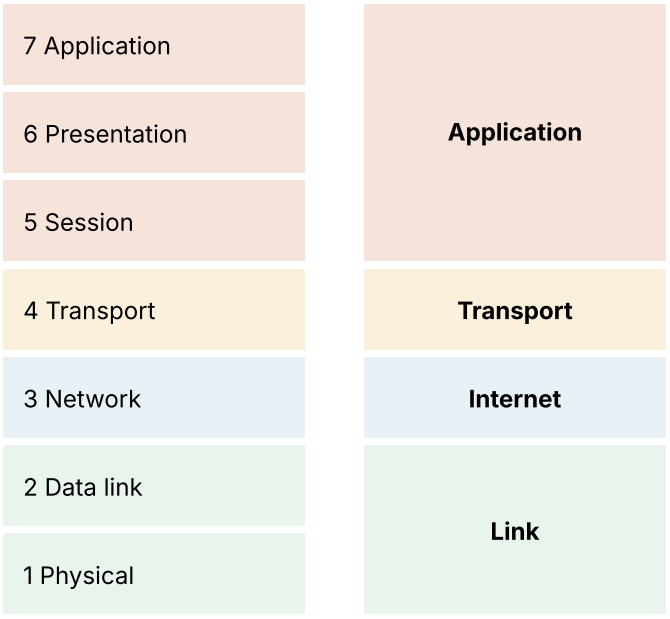
pfSense is the only device plugged into all three segments. So every conversation between zones has to pass through pfSense, where your firewall rules decide what's allowed.

QUICK CHECK (answers at the back)

1. What do DMZ and MGMT stand for?
2. Why is the DMZ given the lowest trust?
3. Why must every conversation between zones go through pfSense?

2 The layers: OSI and TCP/IP

> **Sending data is split into layers. Each layer has one job, and each adds its own label (a header) to the data on the way down.**



OSI (7 layers) TCP/IP (4 layers)

OSI has 7 layers and is used for teaching and troubleshooting. **TCP/IP** has 4 layers and is what the internet actually runs on. They describe the same thing at different levels of detail.

Memory aid (layers 1 up to 7):
Please **D**o **N**ot **T**hrow **S**ausage **P**izza **A**way
Physical, Data link, Network, Transport, Session, Presentation, Application

Layer	Job	Uses	Device	In your lab
7 Application	The actual conversation	HTTP, SSH, DNS	WAF, proxy	Juice Shop, SSH
6 Presentation	Format and encryption	TLS		HTTPS to Wazuh
5 Session	Open and close sessions			Your SSH session
4 Transport	Which program gets the data (ports)	TCP, UDP	Firewall	Port 22, 1514
3 Network	Addresses and routing between networks	IP, ICMP	Router	10.0.x.x, pfSense
2 Data link	Delivery on the local network	Ethernet, MAC, ARP	Switch	VMware segments
1 Physical	Signals on the wire or radio	Cables, Wi-Fi	NIC	Virtual network cards

Encapsulation: wrapping the data

Your SSH attempt from Kali, going down the layers:

```
[ SSH data ]                                          Application
[ TCP: port 56578 → 22 ][ SSH data ]                  Transport
[ IP: 10.0.1.101 → 10.0.2.10 ][ TCP ][ SSH data ]     Internet
[ Ethernet: Kali MAC → pfSense MAC ][ IP ][ TCP ][ data ]   Link
```

pfSense removes the Ethernet label, reads the IP and TCP labels, checks its rules, then wraps the packet in a new Ethernet label for the DMZ. **MAC addresses change at every hop. IP addresses stay the same end to end**, unless NAT rewrites them.

REMEMBER

The layer tells you which log to read. Port scans show in **firewall logs** (layer 4). SQL injection shows in **web server logs** (layer 7). ARP spoofing only shows in **packet captures** (layer 2).

QUICK CHECK (answers at the back)

4. Which TCP/IP layer do port numbers belong to?
5. Which address changes at every hop: MAC or IP?

3 IP addresses and the gateway

> **An IP address is a device's postal address on a network. Part of it names the network (the street) and part names the device (the house number).**

What an IPv4 address looks like

An IPv4 address is **32 bits**, written as **4 numbers from 0 to 255** called **octets** (8 bits each).

```
10      .  0        .  2        .  10
00001010.  00000000 .  00000010 .  00001010
```

Binary in 60 seconds

Each of the 8 bits in an octet has a value. Add up the values of the bits that are 1.

Bit value	128	64	32	16	8	4	2	1
10 in binary	0	0	0	0	1	0	1	0
192 in binary	1	1	0	0	0	0	0	0

10 = 8 + 2. And 192 = 128 + 64. All eight bits on = 255, the largest possible octet.

Private and special addresses

Range	What it is
`10.0.0.0/8`	Private (your lab uses this)
`172.16.0.0/12`	Private
`192.168.0.0/16`	Private (most home routers)
`127.0.0.0/8`	Loopback: the machine talking to itself
`169.254.0.0/16`	Link-local: the machine asked for DHCP and got no answer

Private addresses can't travel on the internet, so a router must translate them (NAT, chapter 8).

The default gateway

When a device wants to reach an address **outside its own network**, it sends the traffic to its **default gateway**: the router's address on its network. In your lab every zone's gateway is pfSense:

Zone	Gateway
LAN	10.0.1.1
DMZ	10.0.2.1
MGMT	10.0.3.1

Static vs dynamic addresses

Servers get static IPs that never change, so everything can find them (Ubuntu 10.0.2.10, Wazuh 10.0.3.10). **User devices** usually get addresses automatically from DHCP (chapter 10).

IN YOUR LAB

Kali is 10.0.1.101 and Ubuntu is 10.0.2.10. Different third numbers under /24 mean **different networks**. So Kali can't send to Ubuntu directly. It sends to its gateway, pfSense, which routes the traffic and filters it with your rules.

QUICK CHECK (answers at the back)

6. What is 10.0.3.10 in binary? (Just the last octet: 10.)
7. A laptop shows 169.254.33.7. What went wrong?
8. What is Ubuntu's default gateway?

4 Subnetting, step by step

> **Subnetting means deciding how many of the 32 bits name the network and how many are left for devices. The /number tells you.**

The /prefix (CIDR notation)

`10.0.2.0/24` means the **first 24 bits are the network** and the remaining **8 bits are for hosts** (devices).

N N N N N N N N N N N N N N N N N N N N N N N N H H H H H H H H

/24: 24 network bits (N), 8 host bits (H). The same split as a **subnet mask** is 255.255.255.0.

The only two formulas you need

$$\text{host bits} = 32 - \text{prefix}$$

e.g. /24 → 32 − 24 = 8 host bits

$$\text{usable hosts} = 2^{\text{host bits}} - 2$$

e.g. $2^8 - 2 = 256 - 2 = 254$

Why − 2? Every subnet reserves two addresses: the **network address** (the first one, all host bits 0), which names the subnet, and the **broadcast address** (the last one, all host bits 1), which reaches every device at once. Neither can be given to a device.

Powers of 2 (learn these)

2^n	2^1	2^2	2^3	2^4	2^5	2^6	2^7	2^8
Value	2	4	8	16	32	64	128	256

The prefix cheat sheet

Prefix	Mask	Host bits	Total addresses	Usable	Block size
/32	255.255.255.255	0	1	1 host*	1
/30	255.255.255.252	2	4	2	4
/29	255.255.255.248	3	8	6	8
/28	255.255.255.240	4	16	14	16
/27	255.255.255.224	5	32	30	32
/26	255.255.255.192	6	64	62	64
/25	255.255.255.128	7	128	126	128
/24	255.255.255.0	8	256	254	256
/16	255.255.0.0	16	65,536	65,534	

*/32 is a single address used in rules to mean "exactly this one machine", like your `Ubuntu_DMZ` alias.

REMEMBER

Total = $2^{\text{host bits}}$. **Usable** = total − 2. Interviewers ask for "usable" on purpose, to see whether you remember the − 2.

4 Subnetting: the method and worked examples

Method A: "I have a prefix. What does the subnet contain?"

1. **Block size** = 256 − the last mask number that isn't 255. (/26: mask .192, so 256 − 192 = 64.)
2. Subnets start at **multiples of the block size**: .0, .64, .128, .192.
3. Find the block your address falls in. Its first address is the **network**, and the address just before the next block is the **broadcast**.
4. Everything between them is **usable**.

Worked example 1: your DMZ, 10.0.2.0/24

Network	First usable	Last usable	Broadcast	Usable
10.0.2.0	10.0.2.1 (pfSense)	10.0.2.254	10.0.2.255	254

Worked example 2: which subnet is 10.0.3.70/26 in?

Block size 64, so blocks start at .0, .64 and .128. 70 is between 64 and 127, so it's in the **.64 block**. Network **10.0.3.64** · usable **.65 to .126** · broadcast **10.0.3.127**.

Method B: "I need X devices. Which prefix?"

1. Find the smallest power of 2 where **$2^n - 2 \geq X$**. That n is your host bits.
2. **Prefix = 32 − n.**
3. Spare addresses = usable − X.

Worked example 3: 25 servers

Host bits	2^n	Usable (−2)	Fits 25?
4	16	14	No, too small
5	**32**	**30**	**Yes**
6	64	62	Yes, but wasteful

Answer: 32 − 5 = **/27**, with 30 usable and **5 spare**. One possible placement: `10.0.3.32/27`, with usable .33 to .62 and broadcast .63.

Worked example 4: 10 cameras

$2^3 - 2 = 6$ is too small, and $2^4 - 2 = 14$ fits. So **/28**: 16 addresses in total, **14 usable**, 4 spare.

Worked example 5: split MGMT 10.0.3.0/24 into four /26

Subnet	Network	Usable	Broadcast
A	10.0.3.0/26	.1 to .62	.63
B	10.0.3.64/26	.65 to .126	.127
C	10.0.3.128/26	.129 to .190	.191
D	10.0.3.192/26	.193 to .254	.255

Same network or not?

Two devices are on the same network only if their **network bits match**. Under /24, that means the first three numbers must match. 10.0.1.101 and 10.0.2.10 differ in the third number, so they're on **different networks** and need a router between them.

IN YOUR LAB

AWS and Azure reserve **5** addresses in every subnet (not 2), so a /24 there gives 251 usable. Cloud VPCs and VNets are usually /16, split into /24 subnets, just like your lab.

QUICK CHECK (answers at the back)

9. How many usable hosts in a /29?
10. You need 50 devices. Which prefix gives the least waste?
11. What is the broadcast address of 10.0.2.128/25?
12. Is 10.0.3.200 inside 10.0.3.128/26?
13. What subnet mask is /27?
14. Why can't you put a new 25-server subnet at 10.0.1.0/24 in your lab?

5 Ports, TCP and UDP

> **The IP address gets data to the right machine. The port number gets it to the right program on that machine.**

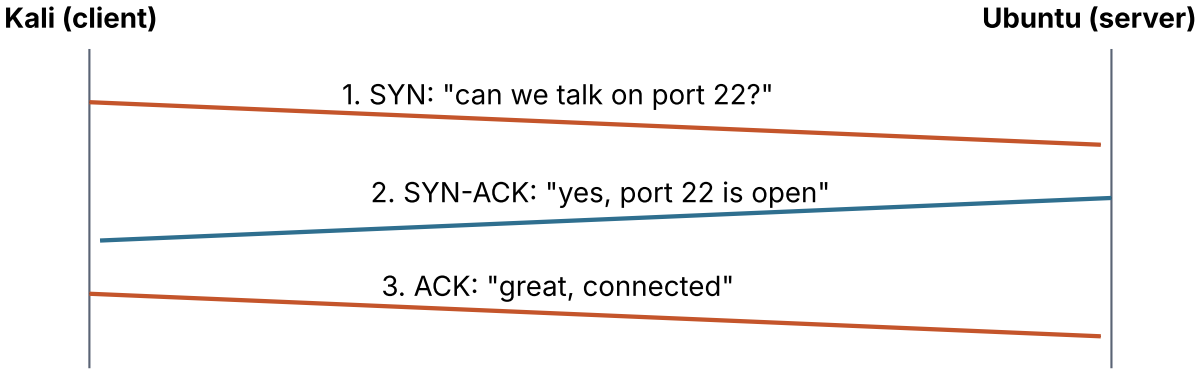
THINK OF IT LIKE THIS

The IP address is the building's street address. The port is the apartment number. Port 22 is "the SSH apartment" and port 443 is "the HTTPS apartment".

TCP vs UDP

	TCP	UDP
Like	A phone call: connect first, every word confirmed	A text message: send it and hope
Connection	Three-way handshake first	None
Delivery	Guaranteed, in order, resent if lost	Best effort, no resend
Speed	Slower	Faster
Used by	SSH, HTTP/HTTPS, Wazuh agent	DNS lookups, DHCP, NTP, video calls

The three-way handshake



Ending: each side sends `FIN` and the other side replies `ACK`. **RST** slams the connection shut instantly, for example when a port is closed.

TCP flags as tcpdump shows them

Shown as	Flag	What it tells an analyst
`[S]`	SYN	Many SYNs to many ports = **port scan**
`[S.]`	SYN-ACK	That port is open
`[.]`	ACK	Normal traffic
`[P.]`	PUSH	Real data flowing
`[F.]`	FIN	Polite close
`[R]`	RST	Port closed, or connection aborted

Port ranges

0–1023: well-known (only root can listen). **1024–49151:** registered. **49152–65535:** ephemeral, the random source ports clients use, like Kali's 56578.

Ports in your lab

Port	Proto	Service	Port	Proto	Service
22	TCP	SSH	1514	TCP	Wazuh agent events
53	UDP	DNS	1515	TCP	Wazuh agent enrolment
80	TCP	HTTP	3000	TCP	Juice Shop
443	TCP	HTTPS (Wazuh dashboard)	4280	TCP	DVWA
123	UDP	NTP	9200 / 55000	TCP	Wazuh indexer / API

Coming with Active Directory: 88 Kerberos, 389 LDAP, 445 SMB, 3389 RDP.

QUICK CHECK (answers at the back)

15. Write the handshake in order.
16. Why does DNS mostly use UDP?
17. You see one IP send [S] to 1,000 ports in 3 seconds. What is it?

6 Switches, routers, MAC and ARP

> **A switch moves data inside one network using MAC addresses. A router moves data between networks using IP addresses.**

Device	OSI layer	Decides using	Keeps a table of	In your lab
Switch	2	MAC address	MAC → port	Each VMware LAN segment
Router	3	IP address	Network → next hop (routing table)	pfSense
Firewall	3–4 (and 7 for next-gen)	IPs, ports, connection state	Active connections (state table)	pfSense, ufw

MAC addresses

Every network card has a **MAC address**, a hardware ID like `00:0c:29:4a:1b:7e`. MACs are used only **on the local network**, and they get replaced at each router hop.

ARP: finding the MAC for an IP

1. Kali wants to send to its gateway 10.0.1.1 but needs its MAC.
2. Kali shouts to everyone on the LAN: *"Who has 10.0.1.1?"*
3. pfSense replies: *"10.0.1.1 is at 00:0c:29:..."*
4. Kali saves the answer in its **ARP cache**. See it with `ip neigh`.

ARP spoofing: an attacker answers "10.0.1.1 is at *my* MAC" and secretly sits in the middle of the conversation. Defences: switch features (Dynamic ARP Inspection) and encrypted protocols like SSH and HTTPS.

Switches and VLANs

A switch learns which MAC is on which port and sends frames only there. A **VLAN** splits one physical switch into several virtual ones that can't talk without a router. Your separate VMware segments give the same effect. **MAC flooding** fills a switch's table so it sends everything everywhere, which lets an attacker eavesdrop.

Routers and the routing table

A router reads the **destination IP**, looks it up in its **routing table**, and forwards the packet. Anything with no specific route goes to the **default route**. On Linux, `ip route` shows it:

```
default via 10.0.1.1 dev eth0
10.0.1.0/24 dev eth0 proto kernel scope link src 10.0.1.101
```

Line 1: send anything unknown to pfSense. Line 2: 10.0.1.x addresses are directly reachable on this card.

QUICK CHECK (answers at the back)

18. Which device uses MAC addresses, and which uses IP addresses?

19. What does ARP do?
20. How does ARP spoofing let an attacker eavesdrop?

7 Firewalls and your rules

> **A firewall allows or blocks traffic based on rules. A stateful firewall also remembers each conversation it allowed, so replies get through automatically.**

Stateless vs stateful

THINK OF IT LIKE THIS

A **stateless** guard has no memory. They need one rule to let your request out and another to let the reply back in. A **stateful** guard keeps a notebook (the **state table**). When the reply arrives, they check the notebook ("Kali started this, so it's expected") and let it through.

Each notebook entry is a **5-tuple**: source IP, source port, destination IP, destination port and protocol.

```
TCP  10.0.1.101:56578  →  10.0.2.10:22   ESTABLISHED
```

The four rules of pfSense rules

1. **Write rules only for who starts the conversation.** Replies are automatic.
2. **Direction is the security.** Kali can connect to Ubuntu, but Ubuntu can't start a connection to Kali. That's what makes a DMZ safe.
3. **Rules are checked on the interface where traffic enters.** Traffic from Kali is judged by the LAN tab, and traffic from Ubuntu by the DMZ tab.
4. **First match wins, then default deny.** Rules are read top to bottom. Anything that matches nothing is dropped.

Your LAN tab, explained

#	Rule	Why it's in this position
1	Allow DNS to pfSense	Everything needs name lookups
2	Allow to Ubuntu on Ubuntu_DMZ_Ports	A specific allow
3	Allow to Wazuh on 1514/1515 and 443	A specific allow
4	**Block Lab_Nets (logged)**	Stops anything else reaching other zones
5	Allow any	What's left is internet traffic only

Swap rules 4 and 5 and "allow any" matches first, so the block never runs. **Order is policy.**

Aliases

An **alias** is a named group of IPs or ports, like `Lab_Nets` or `Ubuntu_DMZ_Ports`. Rules read like plain policy, and a change happens in one place.

Other firewall types

- **Host firewall** protects one machine: ufw on Ubuntu (SSH only from 10.0.1.101).
- **NGFW** (next-generation firewall) also understands apps and users (layer 7).

- **WAF** (web application firewall) blocks web attacks like SQL injection. It belongs in front of Juice Shop.

IN YOUR LAB

Logged block rules turn the firewall into a **sensor**. A hacked machine trying to reach other zones leaves evidence. Later, these logs go to Wazuh. Also remember that **Docker-published ports bypass ufw**, so pfSense is the real control for your web apps.

REMEMBER

AWS **security groups** and Azure **NSGs** are stateful, like pfSense. AWS **network ACLs** are stateless: you must allow the replies too, on ephemeral ports 1024–65535.

QUICK CHECK (answers at the back)

21. Why don't you need a DMZ rule for Ubuntu's SSH replies to Kali?
22. What happens to traffic that matches no rule?
23. Which tab's rules judge traffic coming from Ubuntu?

8 NAT: Network Address Translation

> **NAT rewrites addresses as traffic passes a router. It's how private 10.x addresses get onto the internet.**

- **Outbound NAT (source NAT):** when Ubuntu downloads an update, pfSense swaps the source 10.0.2.10 for its own WAN address (192.168.155.137). It remembers the swap so the reply finds its way back. VMware then does the same thing again on your laptop.
- **Port forwarding (destination NAT):** sends traffic arriving on a public port to a machine inside. **Your lab has none on purpose**, so nothing inside is reachable from outside.
- **NAT is not a security control.** It hides addresses, but firewall rules are what actually block traffic.

THINK OF IT LIKE THIS

A company receptionist sends all outgoing letters with the company's address, not each employee's desk number. When replies come back, they know which desk each one belongs to.

9 DNS: Domain Name System

> **DNS is the internet's phone book. It turns names like packages.wazuh.com into IP addresses. It uses port 53 (UDP, or TCP for big replies).**

How a lookup works

1. Your machine checks its own **cache**.
2. It asks its **resolver**. In your lab that's pfSense (10.0.x.1).
3. If pfSense doesn't know, it asks a **root server**, which points to the **.com servers**, which point to **wazuh.com's own server**.
4. The answer comes back and is **cached** for its TTL (time to live).

Record	Holds	Record	Holds
A	IPv4 address	MX	Mail server
AAAA	IPv6 address	TXT	Text: SPF and DKIM email security
CNAME	Alias to another name	PTR	Reverse lookup: IP → name

Commands: `dig packages.wazuh.com`, `dig -x 10.0.3.10` (reverse), `nslookup`.

Attacks: DNS spoofing or cache poisoning (fake answers), **DNS tunnelling** (hiding stolen data inside lookups), and look-alike phishing domains. **SOC signs:** very long random subdomains, or one host making huge numbers of lookups.

IN YOUR LAB

Every zone may only use pfSense for DNS. Forcing DNS through one resolver means it can be logged and filtered. Your earlier DMZ rule allowed "any" port to pfSense, which let the DMZ reach the pfSense web page. Fixing it to port 53 closed that hole.

QUICK CHECK (answers at the back)

24. Why is NAT not a security control on its own?
25. Which DNS record maps an IP back to a name?
26. Name one sign of DNS tunnelling.

10 DHCP: Dynamic Host Configuration Protocol

> **DHCP hands out IP addresses automatically, along with the gateway, DNS server and subnet mask. It uses UDP ports 67 (server) and 68 (client).**

DORA: the four steps

1. **D**iscover: the new device shouts "is there a DHCP server?"
2. **O**ffer: the server offers an IP.
3. **R**equest: the device asks for that IP.
4. **A**cknowledge: the server confirms, with a lease time.

- **Attacks:** a **rogue DHCP server** hands out a fake gateway to sit in the middle of traffic. **Starvation** uses up every address. Defence: DHCP snooping on switches.
- **SOC value:** DHCP lease logs tell you **which machine had an IP at a given time**. That's vital when an alert names only an IP.

IN YOUR LAB

Your servers use static IPs. Check whether Kali's 10.0.1.101 is a DHCP lease in pfSense: **Status → DHCP Leases**.

11 NTP: Network Time Protocol

> **NTP keeps every clock in agreement. It uses UDP port 123. In security, accurate time is not optional.**

- Time flows down **strata**. Stratum 0 is an atomic or GPS clock, stratum 1 servers connect to it, and each hop adds 1.
- **Best practice:** one internal server (pfSense) syncs from the internet, and every other machine syncs from it.

Why time is a security control

Reason	What breaks with bad time
Correlation	The attack timeline across systems is in the wrong order
Detection	Rules like "8 failed logins in 2 minutes" misfire
Evidence	Reports and legal cases can be challenged
Authentication	Kerberos (Active Directory) fails beyond 5 minutes of difference
Compliance	PCI DSS requirement 10.6 requires time sync

IN YOUR LAB

Your MGMT rules block NTP. The Wazuh clock ran 1 hour ahead (and 35 minutes another day), so alerts were stamped in the future and hidden from "Last 1 hour". Windows keeps the hardware clock in

local time (Lagos, UTC+1), and Linux reads it as **UTC**. **UTC is not your local time.** Next session's fix: pfSense as the lab's NTP server.

12 ICMP and ping

> **ICMP carries network status messages. `ping` uses ICMP echo request and reply to ask "are you there?"**

- ICMP is a protocol like TCP and UDP, so it must be **allowed by a rule**. Your DMZ can ping its gateway, but MGMT can't (no rule). That's expected, not broken.
- `traceroute` uses ICMP replies to show every router hop on the path.
- **Attacks:** ping sweeps (finding live hosts) and ICMP tunnelling. Many networks limit ICMP for this reason.
- A failed ping doesn't prove a machine is down. A firewall may just be blocking ICMP. Test the real service instead (`curl`, `nc -zv`).

QUICK CHECK (answers at the back)

27. What does DORA stand for?
28. Give two reasons time sync matters in a SOC.
29. Wazuh can't ping 10.0.3.1. Is the network broken?

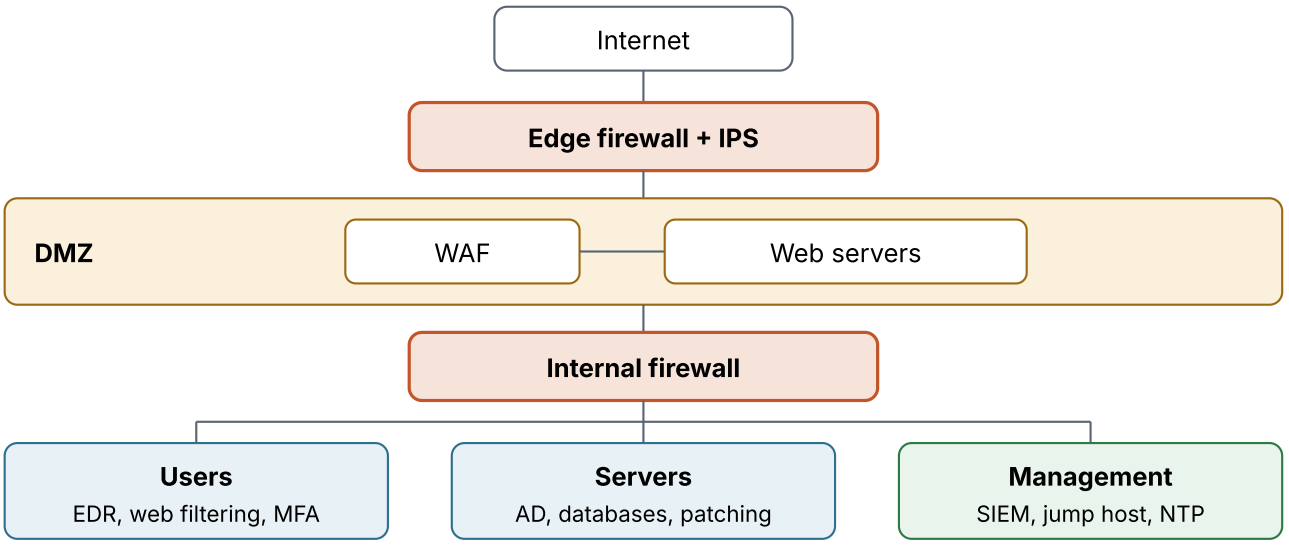
13 Segmentation, micro-segmentation and zero trust

> **Segmentation splits a network into zones by trust, so a breach in one zone can't spread freely. Micro-segmentation applies the same idea to each individual machine.**

Key ideas

- **North-south** traffic goes in and out of the network. **East-west** traffic moves between internal zones, and that's where attackers go after they get in.
- **Lateral movement** (MITRE tactic TA0008) is an attacker hopping from machine to machine. Segmentation is the main defence.
- **Least privilege:** allow only the exact flows each zone needs.
- **Defence in depth:** several layers of control, so one failure isn't a breach. Your lab uses pfSense, ufw, SSH keys and Wazuh.

Where security devices go



Device	Where	Job
Edge firewall	Internet boundary	Allows only published services
IPS / IDS	Inline (IPS) or watching a copy (IDS)	Blocks or spots known attacks. Suricata on pfSense is planned
WAF	In front of web apps	Blocks SQL injection and XSS
Internal firewall	Between the DMZ and internal zones	Contains a hacked DMZ server
Jump host	MGMT	The only way admins reach servers, so all admin access is logged
EDR	Every endpoint	Catches malicious processes
SIEM	MGMT	Collects logs from every zone (Wazuh, later Sentinel)

Micro-segmentation and zero trust

- **Micro-segmentation** applies policy per machine or workload. Your ufw rule (SSH only from Kali) is an example, even inside the DMZ.
- **In the cloud** this is normal: an AWS security group or Azure NSG is attached to each VM.
- **Zero trust:** never trust a connection because of where it comes from. Verify identity, device and context on every request.

QUICK CHECK (answers at the back)

30. What's the difference between north-south and east-west traffic?
31. Where does a WAF go, and why?
32. Give one micro-segmentation example from your lab.

14 Troubleshooting bottom-up

> **Start at the bottom layer and work up. A fault low down breaks everything above it, so going upwards finds the real cause fastest.**

#	Question	Command	From your lab
1	Is the card up, on the right segment?	`ip -br a`	Wazuh on soc-mgmt
2	Right IP and gateway?	`ip a`, `ip route`	Ubuntu static via netplan
3	Can I reach the gateway?	`ping -c 3 10.0.2.1`	MGMT ping fails by design
4	Does DNS work?	`dig packages.wazuh.com`	DMZ DNS rule fixed to 53
5	Is the port listening?	`sudo ss -tlnp`	Wazuh API on 55000
6	Is a firewall blocking it?	pfSense firewall logs	Logged Lab_Nets blocks
7	Does the service work?	`curl -I`, `systemctl status`	A 403 still proved connectivity

REMEMBER

Before every command, **read your prompt**: `user@hostname` tells you which machine you're on. And **test a hypothesis before changing anything**.

15 From your lab to the cloud

> **Everything you've built has a direct cloud equivalent. Cloud security is these same fundamentals, as managed services.**

In your lab	AWS	Azure
LAN, DMZ, MGMT zones	VPC with subnets	VNet with subnets
pfSense rules	Security groups (stateful), network ACLs (stateless)	NSGs (stateful)
Aliases	Prefix lists, security group references	Application security groups
Outbound NAT	NAT gateway	NAT gateway
pfSense DNS	Route 53 Resolver	Azure DNS
SSH keys	EC2 key pairs, Session Manager	SSH keys, Azure Bastion
Wazuh SIEM	Security Lake, partner SIEMs	Microsoft Sentinel
Local accounts	IAM users and roles	Entra ID, Azure RBAC

QUICK CHECK (answers at the back)

33. Which AWS control is stateless?

34. What is the Azure equivalent of your three zones?

16 Glossary

Term	Meaning
Alias	A named group of IPs or ports used in firewall rules
ARP	Finds the MAC address for an IP on the local network
Broadcast address	The last address in a subnet; reaches every device at once
CIDR / prefix	The /number: how many bits are the network part
Default deny	Anything not explicitly allowed is blocked
Default gateway	The router a device uses to reach other networks
DMZ	Demilitarised Zone: a buffer for exposed services
Encapsulation	Each layer wrapping the data in its own header
Ephemeral port	A random high source port a client uses for one connection
Lateral movement	An attacker hopping between internal machines
MAC address	A network card's hardware address, used on the local network
MGMT	Management zone: monitoring and admin tools
NAT	Rewriting addresses at a router
Network address	The first address in a subnet; names the subnet
Octet	One of the four 8-bit numbers in an IPv4 address
Stateful firewall	Remembers allowed connections and lets replies through
State table	The firewall's list of active connections (5-tuples)
Subnet mask	The prefix written as four numbers, e.g. 255.255.255.0
TTL	Time to live: how long a DNS answer may be cached
UTC	Coordinated Universal Time; servers use it. Lagos = UTC+1
VLAN	A virtual network carved out of one physical switch
Zero trust	Verify every request; trust nothing by location

Answers to the quick checks

1. Demilitarised Zone; Management.
2. It hosts services exposed to attackers, so it's the most likely zone to be compromised.
3. pfSense is the only device connected to all three segments, so it's the only path between them.
4. Transport.
5. The MAC address. The IP address stays the same end to end (unless NAT rewrites it).
6. 00001010 (8 + 2).
7. It asked for a DHCP address and got no reply, so it gave itself a link-local address.
8. 10.0.2.1 (pfSense).
9. 6 ($2^3 - 2$).
10. /26 (62 usable).
11. 10.0.2.255.
12. No. 10.0.3.128/26 covers .128 to .191, so 10.0.3.200 falls in the next block, 10.0.3.192/26.
13. 255.255.255.224.
14. 10.0.1.0/24 is already your LAN. Two networks can't use the same addresses, and a /24 wastes addresses for 25 servers. Use a /27 in a free range.
15. SYN → SYN-ACK → ACK.
16. Lookups are tiny and need to be fast. If one is lost, the client simply asks again.
17. A port scan (reconnaissance, MITRE T1046).
18. A switch uses MAC addresses; a router uses IP addresses.
19. It finds the MAC address that belongs to an IP on the local network.
20. The attacker answers ARP requests with their own MAC, so traffic flows through them first.
21. The firewall is stateful. Kali started the connection, so the replies are matched in the state table.
22. It's dropped by default deny.
23. The DMZ tab, because that's where Ubuntu's traffic enters pfSense.
24. It only hides addresses. Firewall rules are what actually allow or block traffic.
25. PTR.
26. Very long, random-looking subdomains, or a huge number of lookups from one host.
27. Discover, Offer, Request, Acknowledge.
28. Any two of: correlation, time-window detections, evidence, Kerberos, compliance.
29. Not necessarily. MGMT has no ICMP rule, so ping is blocked by design. Test the actual service instead.
30. North-south goes in and out of the network; east-west moves between internal zones.
31. Directly in front of web apps, to block web attacks like SQL injection before they reach the app.
32. ufw on Ubuntu allowing SSH only from 10.0.1.101.
33. Network ACLs.
34. A VNet with three subnets, each protected by an NSG.

REMEMBER

Next lab session: make pfSense the lab's time server (NTP), so the Wazuh clock never drifts again. Re-read chapters 7 and 11 before you start.